Adaptive Reserve Dispatch in a Constructed Indian Microgrid: A Reproducible Negative-Result Study

rishav76dev

Abstract

We evaluate forecast-derived battery reserve guidance in a constructed islanded microgrid combining Jaisalmer weather with a scaled Indian national-demand proxy. The three-year input is split chronologically; a validation-selected design is frozen before 3,947 hours of final conventional dispatch. Rule-based and perfect-foresight control both yield 527.693 kWh unserved, while the proposed adaptive-reserve MPC yields 1,692.217 kWh. Marginal interval correction improves nominal 80% interval coverage, but scenario tail coverage remains inadequate. A published-parameter LFP model is numerically checked against its reference implementation. These experiments do not establish an adaptive reliability or pack-lifetime advantage. Source provenance and cell-to-pack applicability remain limitations; this is an exploratory report, not a submission-ready field study.

1 Research question and status

The question is whether synchronized forecast-error trajectories provide useful energy-reserve guidance and a better reliability–degradation trade-off. An advantage is a hypothesis, not an assumption. The original synthetic results remain separately recorded in the repository. This report covers the later constructed Indian reference case and explicitly retains unfavorable outcomes. A comprehensive, independently verified novelty review is not claimed.

2 Data and provenance

We use NASA POWER hourly gridded irradiance, 2-m air temperature and 10-m wind at approximately 26.9157° N, 70.9083° E[3]. The downloaded 2020–2024 weather record contains 43,848 continuous hourly rows. Demand is a pinned secondary mirror of a dataset attributed to Kumar, Mukherjee and Kalita[2, 4]. The mirror reports CC BY 4.0, national MW demand and IST timestamps. The primary Mendeley record was inaccessible through the execution proxy; these claims have not been independently verified against it. The resulting series must not be represented as measured local microgrid load.

The experiment uses 2022–2024: 26,304 hourly observations. Six implausible demand readings are causally forward-filled, with runs limited to three hours. National demand is scaled to a 12 kW training mean. Hour-start IST demand intervals are aligned with UTC weather using equal overlap of neighboring weather hours under a piecewise-constant weather assumption. Simplified PV and wind conversions and training-only reference sizing are documented in the source audit. The 10-m wind curve is not a validated hub-height turbine model.

The split is 60% training, 15% validation, 10% calibration and 15% test. All sizing and fixed-reserve selection use validation only. The final episode contains 3,947 hours; it covers part of one year rather than an independent multi-year site trial. Source hashes, code hashes and configuration are retained.

3 Physics and decision methods

Battery power is positive on discharge. The AC balance is

$$P_{PV} + P_{wind} + P_b + P_{unserved} = P_{load} + P_{curtailed}. \tag{1}$$

The primary comparison uses constant 95% charging/discharging efficiencies, SOC limits 0.1–0.9, initial SOC 0.5 and fixed SOH. Validation exploration found none of the initial modest-renewable designs adequate. The

expanded search selected 200 kWh/50 kW and three times the training-derived renewable capacities, using a declared lexicographic choice rather than an economic optimum.

The backward recursion returns minimum initial energy for each synchronized 24-hour scenario. Power or energy infeasibility remains infinite; the empirical 95th percentile is taken over all outcomes. Infeasible targets remain reported. The primary soft fallback is maximum usable energy; removing infeasible soft targets is a separately labelled counterfactual, not a feasibility claim.

We compare rule based, deterministic MPC, validation-tuned fixed reserve, adaptive reserve, independent-residual reserve, and scenario MPC. Scenario MPC shares its first action but permits optimistic scenario-specific later recourse; it is not a fully non-anticipative multistage tree. Perfect foresight is a non-deployable EENS-only LP over the complete episode.

4 Forecast uncertainty

Nine horizon-aware LightGBM quantile models produce P10/P50/P90 estimates. An interval-widening correction is fitted on early validation and reported on later data. On the final reference test, pooled raw nominal-80% coverage is 74.66%, versus 84.11% after correction. This is empirical coverage under temporal dependence, not an exchangeable conformal guarantee.

The median-residual sampler uses 109 completed joint daily blocks. Its one-sided 95% net-energy bound covers only 75.61% of the evaluated observed 24-hour net energies. Correcting marginal intervals does not recalibrate this sampler. The initial reserve is infeasible at 34.08% of test origins. Residual sampling is not conditioned on origin hour, and the current sampler does not impose an astronomical PV night mask; these are further physical/statistical limitations.

5 Ageing and learning

A separate evaluator implements the BLAST-Lite 250 Ah LFP-Gr model[1]. The published-model parameters and state updates are checked numerically against a pinned upstream reference. The assumed battery temperature is 25°C; 35°C is a sensitivity case. Daily cycle depths outside the experimental range are extrapolations. Numerical agreement with source code is not empirical validation of the intended pack, and model capacity loss is not a measured lifetime extension.

The four SAC variants use identical physics and chronology, three seeds and 10,000/30,000-step candidate budgets. Checkpoints are selected by validation EENS, then modelled damage. The reward uses a published-model-derived hourly damage proxy; daily offline ageing is reported separately. Ageing states are included in the observation. The constrained variant uses an episode-wise dual penalty; physical projection remains hard, but the learned reliability target is not guaranteed. The nested policy-training suffix is only part of the outer training period and does not cover all seasons. Budget comparisons do not establish convergence. Conventional test outcomes were available during later reporting; RL checkpoint selection uses validation only.

6 Results and uncertainty

Rule-based energy LPSP is 1.0554%, narrowly missing the provisional 1% target; perfect foresight gives the same EENS. Thus the target is infeasible even under future knowledge for this selected design and realized episode. No final-test retuning is performed to conceal this outcome.

Adaptive-minus-fixed mean hourly unserved-power difference is 0.07456 kW. A paired circular bootstrap gives a 95% interval [0.00879, 0.13231] kW for 24-hour blocks, but [–0.02759, 0.15614] kW for 168-hour blocks. The latter spans zero; conclusions depend on temporal dependence assumptions. Across the initial conventional controllers, throughput and published-model damage have weak rank association. Serving less load is not evidence of a degradation benefit.

A post-hoc coupled-capacity sensitivity of rule-based dispatch gives 599.586 kWh unserved at 25°C and 642.530 kWh at 35°C, with modelled final SOH 0.98699 and 0.98085. Capacity is updated daily, with removed stored energy recorded separately. This is not the primary fixed-capacity comparison or an RL claim.

controller	eens_kwh	lpsp_energy	published_loss_fraction
rule_based	527.692790	0.010554	0.013006
deterministic_mpc	2264.200549	0.045283	0.010818
tuned_fixed_mpc	1397.912212	0.027957	0.011929
adaptive_max_fallback	1692.217460	0.033843	0.012512
adaptive_no_infeasible_target	1735.335871	0.034706	0.012346
independent_reserve_mpc	2638.939956	0.052777	0.011352
scenario_mpc	1337.342860	0.026746	0.010951
perfect_foresight_non_deployable	527.692790	0.010554	0.009883
sac_vanilla_42	601.704110	0.012034	0.012921
sac_vanilla_43	749.996275	0.014999	0.012785
sac_vanilla_44	537.032533	0.010740	0.012861
sac_fixed_42	532.366751	0.010647	0.012847
sac_fixed_43	2018.556386	0.040370	0.012658
sac_fixed_44	679.986012	0.013599	0.012918
sac_adaptive_42	539.868884	0.010797	0.012823
sac_adaptive_43	546.082899	0.010921	0.012848
sac_adaptive_44	727.233558	0.014544	0.012955
sac_constrained_42	539.955117	0.010799	0.012831
sac_constrained_43	546.759210	0.010935	0.012856
sac_constrained_44	1623.472577	0.032468	0.012827

Table 1: Frozen conventional comparison and validation-selected SAC checkpoints. Capacity loss is a model output, not pack-validated degradation.

7 Structural interpretation and limitations

For the implemented single-bus, fixed-efficiency, fixed-capacity model with inelastic demand, unweighted EENS and a minimum-only terminal constraint, work-conserving greedy dispatch is EENS-optimal. Express stored energy in deliverable AC units: surplus is an arrival to a bounded buffer and served demand is a departure. Deferring feasible service cannot increase total departures; serving it earlier frees room for intervening arrivals. Admitting unused surplus has no holding cost in this objective. Repeated exchanges produce a greedy schedule without reducing total served energy. The repository provides the full scope statement and randomized LP checks.

This does not establish greedy optimality for ageing, critical-load weighting, capacity fade or temperature-dependent physics. A future contribution must demonstrate damage reductions at matched reliability or justify different application constraints. Source-primary verification, seasonal scenario calibration, pack-level validation and robust policy convergence remain necessary before a research submission. The current result is a reproducible negative finding, not demonstrated adaptive superiority.

References

[1] Paul Gasper, Nina Prakash, and Kandler Smith. BLAST-Lite: Battery lifetime analysis and simulation tool-suite. Pinned commit b093495b47dc40dd96dba865d91f553619501e94, model lfp_gr_250AhPrismatic_2019.py; experimental data DOI 10.1016/j.est.2023.109042 cited by upstream. BSD licence retained.

[2] S. Kumar, D. Mukherjee, and K. Kalita. Electricity demand, solar and wind generation data (september 2021–june 2025) of india at 1-hour interval, 2025. Mendeley Data V2. Citation and CC BY 4.0 attribution reported by the secondary mirror; primary metadata was not independently accessed.

[3] NASA POWER Project. Prediction of worldwide energy resources: Hourly data services. Jaisalmer point weather, UTC, 2020–2024; request URLs and hashes retained.

[4] RudranshVyas. India grid demand and renewable generation forecasting. Secondary hourly-data mirror and data dictionary, pinned commit 9fc756aeceefb2e53d6494d48722f5dd771996e0. No code from this repository is incorporated.

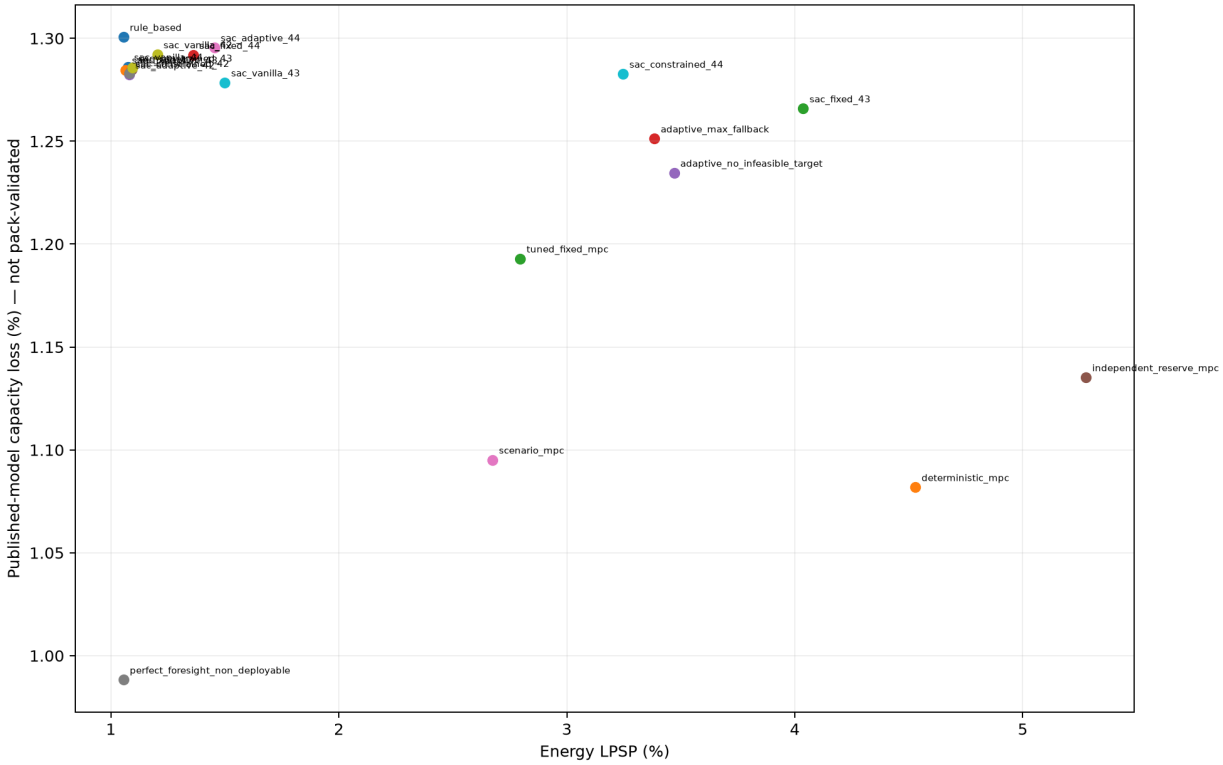


Figure 1: Reliability versus published-model capacity loss. Comparisons at unequal reliability cannot establish an adaptive ageing advantage.